

YIRAN HU

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SUMMARY

Berkeley **Math & CS** sophomore (3.97 GPA); **ICPC** team, USACO Platinum. CTO of Ambees; builds full-stack, iOS, and **LLM eval** systems.

EDUCATION

University of California, Berkeley

Expected May 2029

B.A. in Mathematics & Computer Science, GPA: 3.97

Berkeley, CA

Awards: UC Berkeley ICPC Team (2026), USACO Platinum, USAMO Qualifier, USAMO 2026 Grader, John Suydam Math Prize

Coursework: Data Structures, AI, Discrete Math & Probability, Probability & Random Processes, Linear Algebra, Real Analysis

EXPERIENCE

Co-Founder & CTO

Aug 2025 – Present

Ambees | *early-career recruiting marketplace*

Berkeley, CA

- Designed a **RAG** system and **AI agents** that vet candidates and match them to jobs (**Next.js**, **TypeScript**, **PostgreSQL**)
- Built an **AI voice onboarding agent** on **OpenAI Realtime** that feeds the multi-stage **matching pipeline**
- Raised **\$100K pre-seed** funding after completing Georgetown's **Summer Launch Incubator**

RESEARCH

Data Discovery Program | LLM Tutors for All, UC Berkeley Data Science Undergraduate Studies

Sep 2026 – Present

Project Lead: Edwin Vargas Navarro

Berkeley, CA

- Designing **LLM-as-a-judge** scoring for Berkeley course tutors (**Data 8**, **Data 100**, **E127**), focusing on **self-judge** bias
- Measuring judge-human agreement with **Cohen's kappa** on **OpenRouter** and **NRP** models (**Qwen3**, **gpt-oss**)
- Writing **Python** automated checks that flag answer leakage and missed student mistakes in tutor responses

Undergraduate Research Apprentice | UC Berkeley School of Education

Sep 2025 – Present

Mentor: Prof. Jose Eos Trinidad | Donor-funded \$3,500 summer research stipend

Berkeley, CA

- Co-authoring an *Organization Theory* paper on **AI agents** as tools, assistants, co-workers, and supervisors
- Coauthored a **27-page policy paper** on how **12** large U.S. school districts govern **generative AI**

Data Discovery Program | UC Berkeley Data Science Undergraduate Studies

Jan 2026 – May 2026

Advisor: Eric Van Dusen

Berkeley, CA

- Built **Jupyter** pipelines benchmarking **Qwen2.5** and **TinyLlama** on accuracy, bias, and compute (**Transformers**, **llama.cpp**)

Research Assistant | UC Berkeley Department of Mathematics

Jan 2026 – May 2026

Advisor: Prof. Alan Hammond

Berkeley, CA

- Investigated **Nash equilibria** in **Brownian Boost** tug-of-war games with **resource allocation** dynamics
- Built a web version of the finite game (**tlp-game.vercel.app**); compared human play traces to **ABMN** equilibria

PERSONAL PROJECTS & HACKATHONS

Rubik's Cube Solver | *C++, Group Theory, Permutation Groups, IDA* Search, BFS*

- Wrote a research paper at **SUMaC** (Stanford) on how **group theory** (commutators, conjugation) builds solving methods
- Built a **2x2** solver with **BFS** and hashing, then a **3x3** solver in **C++** with **Kociemba's two-phase** algorithm
- Searched **cosets** of subgroup G_1 with **IDA***, coordinate move tables, and 5 admissible **pruning tables** built by **BFS**
- Solves **1,000** random cubes in **20.5 moves** on average at **3.2 ms** each; verified by **4.8M** unit checks

Kalshi Tennis Trading Bot | *Python, Markov Chains, Kalshi API, ATP Tennis API, Real-time Data*

- Built a live trading bot for ATP tennis contracts on **Kalshi** with a **hierarchical Markov model** (point, game, set, match)
- Added **leverage** and **EWMA momentum** signals with regime-based **maker-only** quoting and per-match inventory caps

bloop | *Swift, SwiftUI, HealthKit, Google Gemini API, WidgetKit* | **HoyaHacks 3rd Place + Best Domain (bloopy.tech)**

- Built a privacy-first **iOS** app on **HealthKit** that tracks headphone sound exposure to protect children's hearing health
- Integrated a **Gemini** agent for listening-pattern analysis; added dose visualization, alerts, Live Activities, and **widgets**

LEADERSHIP & ACTIVITIES

Senior Mentor, CS 88 Course Staff | *Computer Science Mentors (CSM), UC Berkeley*

Sep 2025 – Present

- Lead weekly small-group review sessions on **Python**, **recursion**, and **data abstraction** for CS 88 students
- Created **video walkthroughs** of midterm and final exams for **3+ semesters**, gaining **1,000+ views** from students

SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, Swift, SQL, $\text{\LaTeX}$

ML / AI: PyTorch, NumPy, HuggingFace Transformers, llama.cpp, RAG, LLM evaluation, OpenAI / Gemini / Groq APIs

Tools: Claude Code, MCP (Model Context Protocol), Git/GitHub, Jupyter, PostgreSQL (Neon), Vercel, Xcode, OpenRouter